

an SSD1306 OLED display (OLED 1), an MPU6050 IMU sensor (S1), a VL53L0X Time-of-Flight sensor (S2), a DS18B20 temperature sensor (S3), and a few additional components. The Arduino Nano interfaces with three sensors and an OLED display to measure track level, gauge, and temperature. The design is compact and suitable for portable field use.

A 3.7V rechargeable Li-ion/LiPo battery powers the system. Since the Arduino Nano and sensors require a regulated 5V supply, the battery output is stepped up using a DC-DC boost converter. The regulated 5V output is connected to the Arduino Nano and all peripheral modules. All sensors, except the DS18B20, communicate with the Nano through the I²C bus using the SDA

and SCL lines.

The MPU6050, VL53L0X, and OLED display communicate using the I²C protocol. This allows multiple devices to share the same two signal lines, resulting in a clean and simplified wiring layout. A single push-button (SW1) connected to digital pin 8 (D5) is used to switch between different display modes.

On the first press of SW1, the display shows the tilt or angle data from the MPU6050. The second press displays the gauge or distance measured by the VL53L0X sensor, while the third press shows the temperature reading from the DS18B20 sensor. With each subsequent press, the cycle repeats continuously.

The TP4056 module's output is connected to the Arduino Nano's

Vin pin. The Arduino Nano 5V supply powers the OLED display, MPU6050, and VL53L0X sensors. A common ground is maintained for all modules to ensure stable operation.

A 0.96-inch (2.4cm) or 1.3-inch (3.3cm) I²C-based OLED display using the SSD1306 controller is used to display the measured parameters. Depending on the selected operating mode, the display shows the tilt angle, gauge (distance), or temperature. The OLED is interfaced with the Arduino Nano via the I²C bus, with SDA connected to pin A4 and SCL to pin A5.

The MPU6050 is a combined three-axis accelerometer and three-axis gyroscope module. It is used

to measure the track's tilt and roll angles. In this system, the roll-angle data is utilised to determine the super elevation (cant).

The VL53L0X is a laser-based Time-of-Flight distance sensor capable of measuring distances up to approximately 2 metres with millimetre resolution. It is used to measure the track gauge or gap distance. The sensor operates on the I²C interface and shares the SDA, SCL, V_{CC}, and GND lines with the MPU6050 and OLED display.

The DS18B20 is a digital temperature sensor that uses the 1-Wire communication protocol. It provides temperature measurements from -55°C to +125°C, with a typical accuracy of ±0.5°C. The data pin is connected to Arduino Nano digital pin D2. A 4.7kΩ pull-up resistor is connected between the data line and the 5V supply, as required for 1-Wire operation.

Software

To program the Arduino Nano, install the Arduino IDE on a Windows, macOS, or Linux system. Upload the code using a standard USB cable. Before uploading, install all required libraries via the Arduino Library Manager. These include Wire.h for I²C communication, Adafruit_SSD1306, and Adafruit_GFX for the OLED display, VL53L0X.h for the distance sensor, and either MPU6050.h or Adafruit_MPU6050.h for roll-angle measurement. For temperature measurement using the DS18B20, OneWire.h and DallasTemperature.h are also required.

In the program, the roll-angle offset is computed using a simple method, such as taking the median of multiple sensor readings, to obtain stable, noise-free data from the MPU6050. Once the angle is calculated in radians, it is used to determine the track super elevation. Super elevation (cant) is defined as the height difference between two parallel rails; it helps provide a

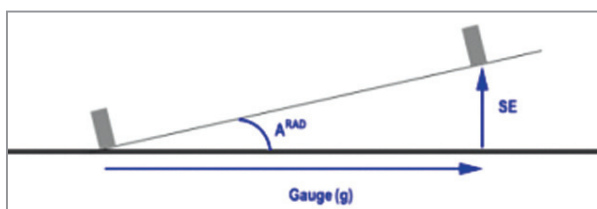


Fig. 3: Angle calculation

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1  #include <Wire.h>
2  #include <VL53L0X.h>
3  #include <Adafruit_SSD1306.h>
4  #include <OneWire.h>
5  #include <DallasTemperature.h>
6
7  // ----- OLED -----
8  #define SCREEN_WIDTH 128
9  #define SCREEN_HEIGHT 32
10 Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);
11
12 // ----- VL53L0X -----
13 VL53L0X sensor;
14 bool vL53Present = false;
15
16 // ----- VL53 smoothing -----
17 const uint8_t VL53_MEDIAN_N = 5;
18 const int VL53_MAX_STEP = 20;
19 const int VL53_JUMP_REJECT = 150;
20 const int VL53_MIN = 30;
21 const int VL53_MAX = 2000;
22
23 int lastValid = -1;
24 float emaDist = -1;
25 const float EMA_ALPHA = 0.25;
26
27 // helper: read raw once
28 int readVL53RawOnce() {
29     int d = sensor.readRangeContinuousMillimeters();
30     if (sensor.timeoutOccurred()) return -1;
31     if (d < VL53_MIN || d > VL53_MAX) return -1;
32     return d;
33 }
34
35 // helper: median of N samples
36 int medianOfSamples() {

```

Fig. 4: Code snippet